

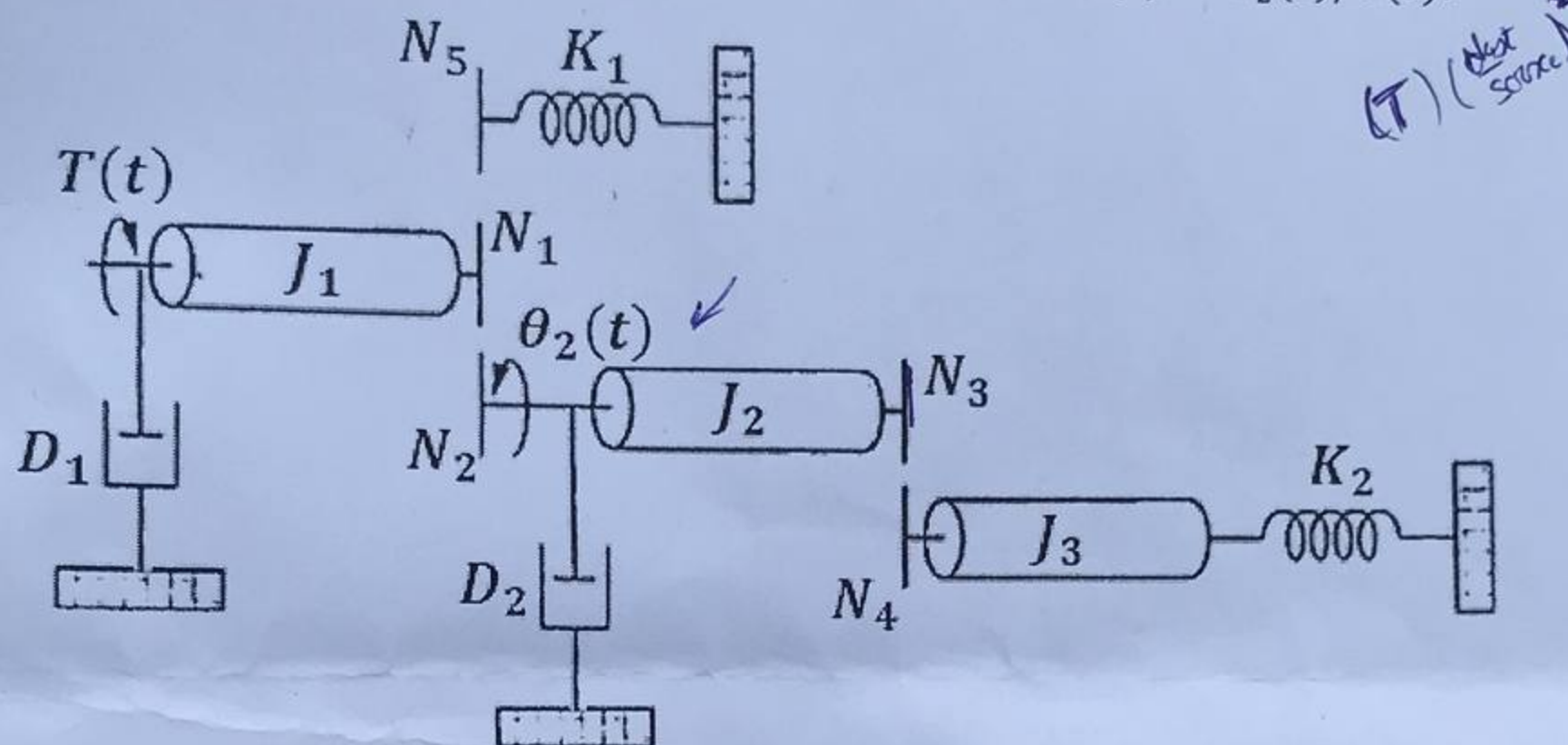


National University of Computer and Emerging Sciences, Lahore Campus

Course:	Feedback Control Systems	Course Code:	EE304
Program:	BS(Electrical Engineering)	Semester:	Fall 2018
Duration:	1 Hour	Total Marks:	30
Date:	2-Oct-18	Weight:	15%
Section:	EE	Page(s):	1
Exam:	MID-1	CLO:	1
Name:		Roll no.:	

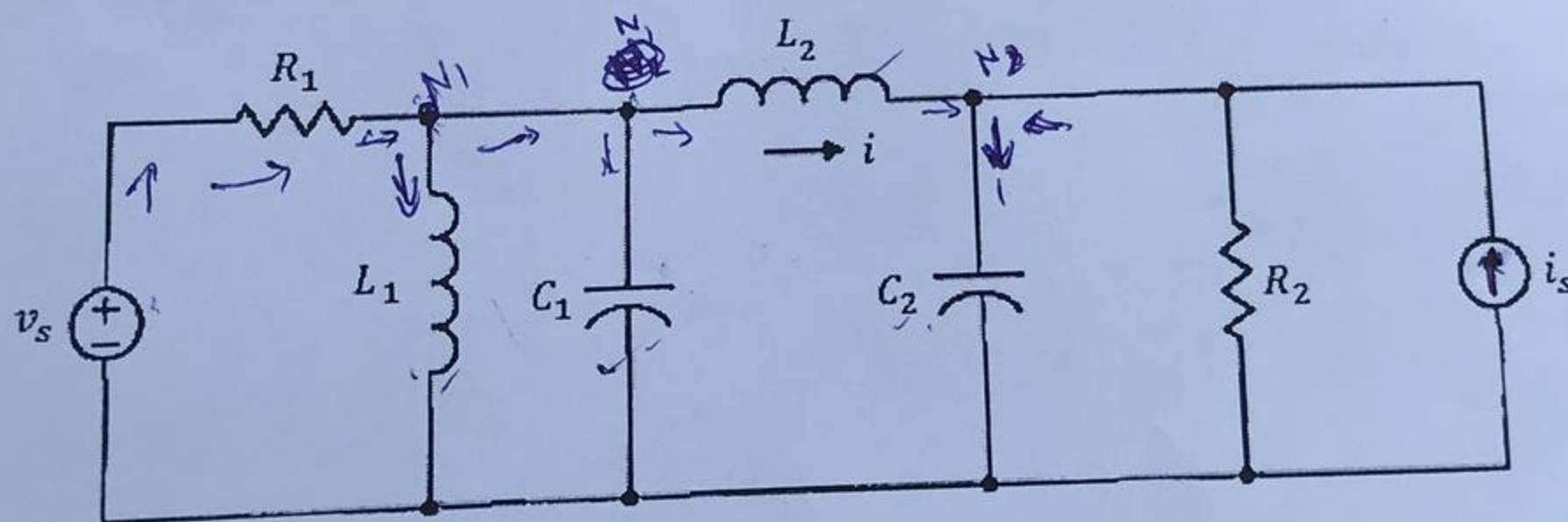
Question # 1

For the rotational system shown below, find the transfer function $G(s) = \theta_2(s)/T(s)$. [15]



Question # 2

Obtain the state space model of the following electrical system. Input variables are v_s, i_s and output variables are v_{R1}, i_{R1}, v_{R2} . Also draw its equivalent block diagram. [15]



$$-v_{C1} - v_s + i_{L1} + C_1 \frac{dv}{dt} + v_{R3} = 0$$

$$\frac{v_{R2}}{C_2} =$$

$$i_L = \frac{C dv}{dt}$$

$$L \frac{di}{dt} = v$$